YEAR 5 MATHEMATICS · UNIT 5 · WORKSHEET

Below Zero in Words

Adding and subtracting across zero: temperatures, floors and money

Mr. David

Name: _____ **Class:** _____

Date given: _____ **Date due:** _____

How to do this worksheet

- Write your answers **on this sheet**.
- In a word problem, **write the number sentence first**, then the answer.
- Stuck? Draw a number line with **0 in the middle**.
- A new word? Look for the blue **Word help** box.

Remember from the lesson

Zero is a bridge. First jump to 0. Then jump again.

3 − 8: 3 to 0 is 3 jumps, then 5 more jumps. 3 − 8 = −5

Section A · Number lines

A1 Count on and count back

Write the missing numbers. The steps are always the same size.

(a) 9, 6, 3, _____, _____, _____

(b) −11, −7, −3, _____, _____, _____

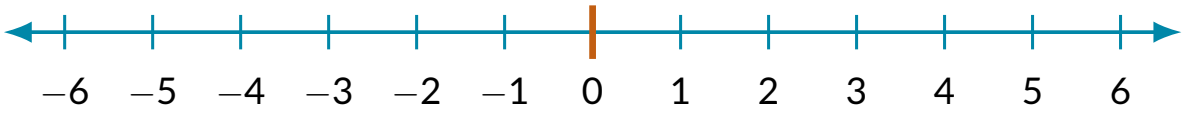
(c) 14, 9, 4, _____, _____, _____

(d) _____, −15, −5, 5, _____, _____

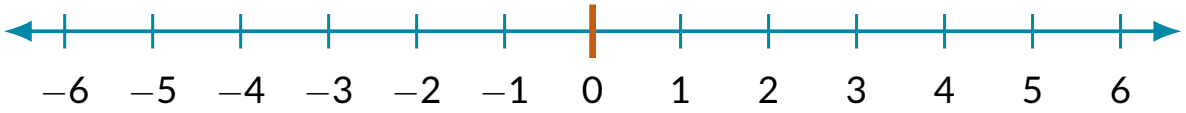
A2 Show the jumps

Draw the jumps on the number line. Then write the answer.

(a) 3 − 7 = _____



(b) −5 + 9 = _____



A3 Work them out

The numbers get bigger as you go. Use zero as a bridge.

(a) 2 − 6 = _____

(b) 1 − 9 = _____

(c) −4 + 7 = _____

(d) −8 + 3 = _____

(e) 5 − 12 = _____

(f) −10 + 16 = _____

(g) 20 − 35 = _____

(h) −25 + 40 = _____

(i) 7 − 100 = _____

(j) −30 + 120 = _____

(k) 15 − 204 = _____

(l) −60 + 1005 = _____

Section B · The English of the question

Word help — these words tell you to add or subtract

rises, **goes up**, **gets warmer**	add (+)
falls, **drops**, **goes down**, **colder**	subtract (−)
deeper	further down: subtract (−)
ground floor (G), **B1**, **B2**	floor 0, then −1, −2 under the ground
owe	you must pay money back. Money you owe is **negative**.
subtract 5 from 2	start at 2: 2 − 5. The number after **from** comes first.
the difference between	the gap between two numbers. It is never negative.

B2 Number sentence first

Write the number sentence. Then write the answer.

1. The temperature is 2 °C. It falls by 9 °C.
   Number sentence: _____ Answer: _____

2. The lift is on floor 3. It goes down 7 floors.
   Number sentence: _____ Floor: _____

3. The lift is on floor B5. It goes up 8 floors.
   Number sentence: _____ Floor: _____

4. Mr. David has 15 000 dong. He buys a book for 40 000 dong. He borrows the rest. How much does he owe?

   Number sentence: _____ Answer: _____

5. Subtract 12 from 5.

   Number sentence: _____ Answer: _____

B3 The difference between

Count the gap. A number line helps.

(a) The difference between 3 and 10. _____

(b) The difference between −2 and 6. _____

(c) The difference between −9 and 4. _____

(d) How much warmer is 8 °C than −5 °C? _____

(e) How many floors is it from floor B3 up to floor 6? _____

Section C · Read the table

C1 One morning in January

The table shows the temperature at 7 a.m. in five places on the same day. (The numbers are made up for this worksheet.)

Place	Temperature
Mộc Châu, Vietnam	4 °C
Hạ Long, Vietnam	13 °C
Beijing, China	−6 °C
Ulaanbaatar, Mongolia	−24 °C
Sydney, Australia	26 °C

(a) Which place is the coldest? _____

(b) Beijing gets 9 °C warmer. What is the new temperature? _____

(c) In Mộc Châu it falls by 7 °C. What is the new temperature? _____

(d) The difference between Sydney and Beijing? _____

(e) Write the five temperatures in order, **coldest first**.

_____, _____, _____, _____, _____

C2 Estimate first

Use easy numbers to estimate. Then work out the exact answer.

Calculation	Estimate	Exact answer
−49 + 151		
302 − 498		
−99 + 1001		

Section D · Think about it

D1 Mr. David's homework

Mr. David got all three wrong. Write the correct answer. Then say what he did wrong.

(a) 6 − 10 = 4 Correct answer: _____

(b) −3 + 8 = −11 Correct answer: _____

(c) The difference between −5 and 3 is 2. Correct answer: _____

D2 Make up the story

Write a word problem for each number sentence. Use a temperature, a lift or money. Use a word from the Word help box. End with a question.

(a) 4 − 9 = −5

(b) −3 + 10 = 7

Answer Key — Teacher Copy

A1 Count on and count back

(a) 0, −3, −6 (b) 1, 5, 9 (c) −1, −6, −11 (d) −25 … 15, 25 (steps of 10). A student who writes −1 after 3 in (a) has stopped seeing the step size at zero: count the jumps aloud with them.

A2 Show the jumps

(a) −4: 3 jumps left to 0, then 4 more. (b) 4: 5 jumps right to 0, then 4 more. Check the arrow starts at the first number, not at 0.

A3 Work them out

(a) −4 (b) −8 (c) 3 (d) −5 (e) −7 (f) 6 (g) −15 (h) 15 (i) −93 (j) 90 (k) −189 (l) 945.
The common wrong answers are the gap with the wrong sign (7 − 100 = 93) and the "keep the minus" sum (−4 + 7 = −11, the scheme of work's misconception). Both mean the student is not starting at the first number.

B2 Number sentence first

1. 2 − 9 = −7 °C 2. 3 − 7 = −4, floor B4 3. −5 + 8 = 3, floor 3 4. 15 000 − 40 000 = −25 000: he owes 25 000 dong 5. 5 − 12 = −7.
In 5, 12 − 5 = 7 is the English mistake (the number after *from* comes first), not an arithmetic one: ask the student to read the sentence aloud.

B3 The difference between

(a) 7 (b) 8 (c) 13 (d) 13 °C (e) 9 floors (3 up to G, then 6). Answers of 4, 5 or 3 in (b)–(e) come from subtracting the digits and ignoring the minus sign. A difference is never negative: −13 is wrong.

C1 One morning in January

(a) Ulaanbaatar (b) −6 + 9 = 3 °C (c) 4 − 7 = −3 °C (d) 26 + 6 = 32 °C (e) −24, −6, 4, 13, 26 (°C). In (e), students who put −6 before −24 are ordering by the digits; ask which is colder.

C2 Estimate first

−50 + 150 = 100, exact 102. 300 − 500 = −200, exact −196. −100 + 1000 = 900, exact 902. Any sensible rounded estimate is fine; the point is that the exact answer is close to it.

D1 Mr. David's homework

(a) −4: he found the gap between 6 and 10 instead of starting at 6 and going down 10. (b) 5: he added 3 and 8 and kept the minus sign. (c) 8: he subtracted 3 from 5; the gap crosses zero, 5 + 3.

D2 Make up the story

Check in this order: (1) it **starts** at the first number (4, or −3); (2) the word matches the sign (falls / goes down / spends for (a), rises / goes up / earns for (b)); (3) full sentences ending in a question; (4) the answer is −5 or 7. Example (a): "It is 4 °C. The temperature falls by 9 °C. What is the temperature now?" Example (b): "The lift is on B3. It goes up 10 floors. Which floor is it on now?"